

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1714

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 2

**Duration : 1 Hour
Total Marks:30**

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I Arsat.P-192424038 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



- iii. Only one employee per task
- iv. E4 cannot handle Records
- v. All employees must be assigned exactly one task

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule.

2. A package-delivery robot navigates a 5x5 city-block grid from Start (S) to a delivery address (G). Certain streets are one-way, restricting movement direction in some cells.

Constraints:

- i. Movement allowed: Up, Down, Left, Right, but one-way cells restrict at least one direction
- ii. Cost of each move = 1
- iii. Cannot move through blocked/no-entry cells in the restricted direction
- iv. Use Manhattan Distance heuristic

Using the above constraints and BFS Search Algorithm, show step-by-step node expansion and frontier updates, and determine the optimal path and cost.

Student 3: ARSAT P | Register Number: 192424038

1. A university must assign 3 professors (P1, P2, P3) to 3 courses (AI, Data Structures, Networks) offered in 3 different time slots such that:

Constraints:

- i. P1 cannot teach in the Networks slot
- ii. P2 must be scheduled before P3 (AI < Data Structures < Networks)
- iii. Only one professor per course/slot
- iv. P3 cannot teach AI
- v. Every professor must be assigned exactly one course

Apply Backtracking Search to find a valid timetable, showing intermediate steps and constraint checks. State the final valid schedule.

2. A cleaning robot operates in a 5x5 room-grid from Start (S) to a docking Goal (G), with a limited battery allowing at most 8 moves.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1 (battery unit)
- iii. Cannot pass through furniture (obstacle) cells
- iv. The robot must reach G within the battery limit

Using the above constraints and Depth-Limited Search (with the battery limit as depth bound), show step-by-step node expansion and determine whether a valid path exists within the limit, along with its cost.

- ii. Tech2 must be assigned before Tech3 (Machine X < Machine Y < Machine Z)
 - iii. Only one technician per machine
 - iv. Tech3 cannot repair Machine X
 - v. All technicians must be assigned exactly one machine
- Apply Backtracking Search to find a valid assignment, showing all intermediate steps and constraint checks. State the final valid schedule.

2. A delivery robot operates in a grid environment (5x5). It must move from Start (S) to Goal (G). Obstacles block certain cells.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles
- iv. Use Manhattan Distance heuristic

Using the above constraints and BFS Search Algorithm, show step-by-step node expansion and frontier updates, and determine the optimal path and cost.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Backtracking search - University timetable

Given:

Professor : P_1, P_2, P_3

Courses (Time slots):

- * AI
- * Data structure
- * Network

Constraints:

- (i) P_1 cannot teach Network
- (ii) P_2 must be scheduled before P_3
- (iii) Only one professor can be assigned to each course
- (iv) P_3 cannot teach AI
- (v) Every professor must be assigned exactly one course

Backtracking steps:

Step 1:- Assign $P_1 \rightarrow AI$

Course	Professor	Status
AI	P_1	Valid
Data structure	-	
Network	-	

Constraint check:-

- * P_1 is not assigned to Network
- * P_3 is not assigned to A1

Step 2:- Try $P_3 \rightarrow$ Data Structure

Course	Professor
A1	P_1
Data Structure	P_3
Network	-

Constraint check:-

- * P_2 must be before P_3

Result: Constraint Violated

Backtrack:- Remove P_3 from Data Structure

Step 3:- Assign $P_2 \rightarrow$ Data Structure

Course	Professor
A1	P_1
DS	P_2
Network	-

Constraint check:-

- * P_2 is before P_3

Step 4:- Assigning remaining professors P_3

Course	Professor
A1	P_1
DS	P_2
Network	P_3

nal Timetable:

Course	professor
AI	P ₁
DS	P ₂
Network	P ₃

2) Depth-Limited Search (Battery limit = 8)

Room Grid:

```

S   .   x   .   .
.   .   x   .   .
.   .   .   x   .
x   x   .   .   .
.   .   .   .   G
  
```

Legend:

* S = Start

* G = Goal

* x = obstacle

* . = Free Cell

Battery limit = 8 moves

Node expansion:

step	Current Node	Depth	Action
1	S(0,0)	0	Start
2	(0,1)	1	Move right
3	(1,1)	2	Move Down
4	(2,1)	3	Move Down
5	(2,2)	4	Move Right

6	(3,2)	5	Move Down
7	(3,3)	6	Move Right
8	(4,3)	7	Move Down
9	G(4,4)	8	Move Right (Goal Reached)

Path Found:

S $\rightarrow$ Right $\rightarrow$ Down $\rightarrow$ ~~Right~~ $\rightarrow$ Down $\rightarrow$ Right $\rightarrow$ Down $\rightarrow$ Right $\rightarrow$ Down $\rightarrow$ Right $\rightarrow$ Down $\rightarrow$ Right $\rightarrow$ Down $\rightarrow$ Right $\rightarrow$ G

Cost Calculation:

* Cost per move = 1 battery unit

* Total move = 8

Total Cost = 8 battery units

Conclusion:

The goal is reached within the battery

limit of 8 moves

Final Answer:

* Valid path exists

* Total path cost = 8

* Goal reached Successfully using Depth-Limited Search.